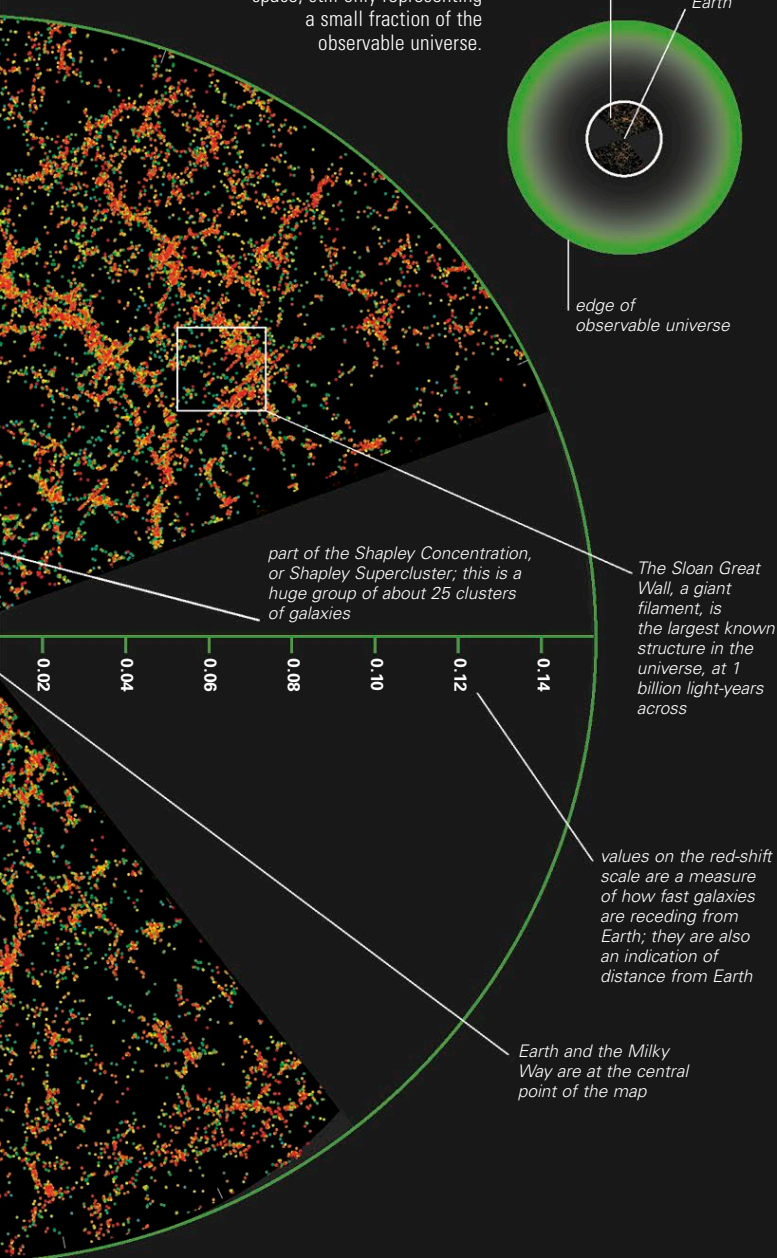


REGION DETAILED BY GALAXY MAP

The galaxy map below covers two thin, wedge-shaped regions of space, still only representing a small fraction of the observable universe.



area depicted in survey galaxy map

Earth

edge of observable universe

part of the Shapley Concentration, or Shapley Supercluster; this is a huge group of about 25 clusters of galaxies

The Sloan Great Wall, a giant filament, is the largest known structure in the universe, at 1 billion light-years across

values on the red-shift scale are a measure of how fast galaxies are receding from Earth; they are also an indication of distance from Earth

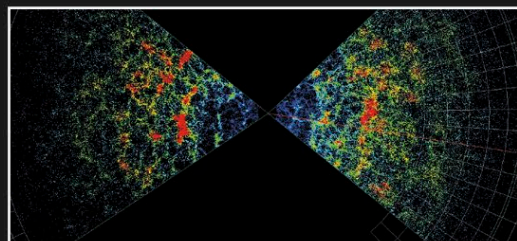
Earth and the Milky Way are at the central point of the map

ACCELERATING EXPANSION

One of the most remarkable recent astronomical discoveries has been the fact that cosmic expansion is accelerating. Studies of type 1a supernovae (see p.283) have revealed that they are unexpectedly faint in the most distant galaxies, which implies that they are farther away than they should be if the rate of expansion of the universe was constant or slowing down. Many cosmologists had expected the expansion of the universe to slow down as the initial impetus from the Big Bang began to fade, so the discovery that its expansion is getting faster implied that an important factor was missing from cosmological theories. Furthermore, the acceleration seems to have begun only around 5 billion years ago, with the universe slowing as predicted until then. Since its discovery in 1998, the accelerating expansion has been corroborated from other measurements, and it is now generally believed to be due to dark energy (see p.58). According to recent measurements, dark energy may be the most abundant form of mass-energy in the universe, accounting for almost 73 percent of the total.

MAPPING DEEP SPACE

While galaxy motions on a local scale are affected by gravitational influences such as the presence of superclusters, on the scale of the universe as a whole, these effects should become negligible in comparison to the overall cosmic expansion resulting from the Big Bang (see pp.48–51). According to Hubble's Law, the speed at which a far-off galaxy is moving away from us is, on average, proportional to its distance, and as a result the red shift in a distant galaxy's light can be used as a measure of its distance. The first large-scale survey of galaxy red shifts, carried out by the Harvard-Smithsonian Center for Astrophysics (CfA), started in 1977 and took 5 years to measure 13,000 galaxies. Since then, other surveys, such as the Sloan Digital Sky Survey and Two-degree-Field Galaxy Redshift Survey (2dFGRS), have mapped many more galaxies. These surveys have confirmed that the large-scale pattern of galaxy distribution remains essentially identical out to distances of billions of light-years.



GALAXY DISTRIBUTION PLOT FROM THE 2DFGRS

Centered on Earth, this plot shows the positions of over 230,000 galaxies. The dots are galaxies and the colors indicate density, with dense regions redder and less dense ones bluer.

EXPLORING SPACE

OBSERVING A MILLION GALAXIES

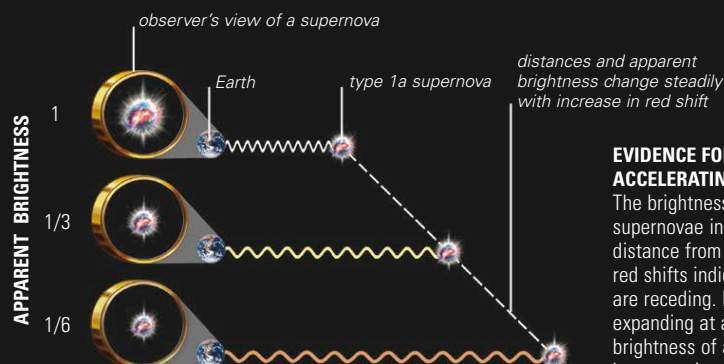
Major galaxy red-shift surveys typically use multi-object spectrographs—devices that can simultaneously record the spectra of hundreds of objects. Instruments such as the Gemini multi-object spectrographs, mounted on two large telescopes in Hawaii and Chile, use special masks to separate light from the different objects before splitting it through a diffraction grating to obtain the spectra.



GEMINI OBSERVATORY

The Gemini observatory has two 26.9-ft (8.1-m) reflectors—one in Chile (shown above), the other in Hawaii—each fitted with a spectrograph for multi-object spectroscopy.

IF EXPANSION HAS OCCURRED AT A STEADY RATE



EVIDENCE FOR ACCELERATING EXPANSION

The brightness of type 1a supernovae indicates their distance from Earth, while their red shifts indicate how fast they are receding. If the universe is expanding at a steady rate, the brightness of a supernova should be proportional to its red shift (top). However, studies have found that distant supernovae are fainter than their red shifts suggest (bottom), indicating that expansion of the universe is accelerating.

IF EXPANSION HAS ACCELERATED

